

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q21-1761

Spencer 138 kV  
June 24, 2023

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1761. The Interconnection Customer proposes a 1026.8 MW natural gas combined cycle facility interconnecting at the Spencer 138 kV substation in the Maryland area. The request was received on May 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 398 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$50,773,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                           | Contingency                          | Loading (%) | Rating (MVA) | DFAX (%) |
|----------------------------------------------|--------------------------------------|-------------|--------------|----------|
| Dobbins / Spencer 138.0 kV line              | Bozman / Spencer 138.0 kV            | 153.1       | 494          | 67.3     |
| Bozman Substation / Spencer 138.0 kV L       | Dobbins / Spencer 138.0 kV line      | 142.3       | 406          | 55.8     |
| Easton / Dobbins Substation 138.0 kV         | Bozman / Spencer 138.0 kV            | 115.8       | 636          | 67.3     |
| Easton / Navy Mole 138.0 kV                  | Bozman / Spencer Substation 138.0 kV | 114.7       | 609          | 67.3     |
| Bozman Substation / Greenstone 138.0 kV Line | Dobbins / Spencer 138.0 kV           | 103.6       | 286          | 28.5     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                       | Required Upgrade                            | Cost Estimate | Allocation to Project |
|--------------------------------|---------------------------------------------|---------------|-----------------------|
| Spencer / Dobbins 138.0 kV     | Rebuild line with higher-capacity conductor | \$6,566,000   | 90% (\$5,912,000)     |
| Bozman / Spencer 138.0 kV Line | Rebuild line with higher-capacity conductor | \$27,494,000  | 100% (\$27,494,000)   |
| Dobbins / Easton 138.0 kV L    | Rebuild line with higher-capacity conductor | \$2,805,000   | 90% (\$2,526,000)     |

| Facility                                     | Required Upgrade                            | Cost Estimate | Allocation to Project |
|----------------------------------------------|---------------------------------------------|---------------|-----------------------|
| Easton / Navy Mole 138.0 kV                  | Rebuild line with higher-capacity conductor | \$8,361,000   | 90% (\$7,528,000)     |
| Bozman Substation / Greenstone 138.0 kV Line | Reconductor line                            | \$7,313,000   | 100% (\$7,313,000)    |
|                                              | Total allocated to project                  |               | \$50,773,000          |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.